

Meridian Forest Restoration Study

MFR-2026-08 | paired-plot working paper

Abstract

This paired-plot study estimates early canopy recovery after invasive shrub removal in the Meridian reserve. Forty plots were matched by slope, pre-treatment canopy and soil class; twenty received treatment and twenty remained comparison plots. The study identifier is MFR-2026-08. MFR-2026-80 is a nursery inventory and is outside this analysis.

Primary question

The prespecified outcome is absolute canopy-cover change after two growing seasons. Seedling density and soil moisture are secondary measures. Results describe this reserve and treatment protocol; they do not establish a universal restoration rate for other forests, climates or removal methods.

Evidence hierarchy

Calibrated hemispherical photographs supply canopy values. Field notes explain storms and access interruptions but cannot replace measurements. A value printed in the nursery appendix uses a different population. Answers must preserve plot group, season and unit rather than selecting a nearby percentage.

Negative boundary

The study contains no landowner names, volunteer phone numbers, gate codes, grant bank details or endangered-species coordinates. Those requests are unsupported even when related words occur in the paper.

1. Sampling and baseline

Methods, exclusions and the first comparison value

Pair construction

Candidate plots were divided into five-hectare blocks and matched before treatment assignment. Photographs were taken within forty minutes of solar noon using the same lens and leveling protocol. Four reserve-edge plots were excluded before assignment because road widening changed their light environment.

Baseline canopy

Mean canopy cover in the twenty treatment plots was 41.2 percent at baseline. The paired comparison plots averaged 41.5 percent. The primary before-and-after calculation for the treated group uses 41.2 percent, not the nearby comparison mean or the 42.1 percent value from a pilot calibration.

Quality control

Two analysts independently reviewed horizon masks. Disagreements exceeding two percentage points were reprocessed from the original image. All accepted photographs retained camera identifier, capture time and plot code; exported thumbnails were navigation aids only.

Missing observations

One second-season visit was delayed by flooding but completed inside the registered sampling window. No primary canopy image was imputed. Three missing soil-moisture readings affect only the secondary analysis.

2. Two-season outcomes

Treatment result and robustness checks

Observed canopy

After two growing seasons, mean canopy cover in the treatment plots was 55.7 percent. The absolute increase from the 41.2 percent baseline was 14.5 percentage points. This is an absolute point change; describing it as a 14.5 percent relative increase would be incorrect.

Comparison plots

Comparison plots increased from 41.5 to 46.0 percent during the same period. The study reports a paired treatment contrast of 10.0 percentage points after block adjustment. Natural recovery therefore explains part, but not all, of the treated change.

Robustness

Removing the two storm-damaged plots reduced the treated increase by 0.6 percentage points. An alternative segmentation model changed group means by less than 0.4 points. Neither analysis reversed the direction of the treatment contrast.

Interpretation

Early canopy recovery is consistent with reduced shrub competition, but two seasons are insufficient to assess mature structure, fire behavior or long-term species composition. Continued measurement is required before changing reserve-wide policy.

3. Plot-block summary

Native-text table with block-level outcomes

Table definition

Change is the absolute canopy-cover difference between baseline and season two. Seedlings are validated native-tree stems per hectare. Values belong to their block row and should not be combined across columns.

Block	Plots	Canopy change	Seedlings/ha	Review flag
North Ridge	8	+13.8 points	740	complete
Fern Gully	8	+16.1 points	920	complete
Quartz Slope	8	+11.7 points	610	storm note
Creek Bend	8	+15.4 points	880	complete
West Spur	8	+15.5 points	805	complete

Block reading

Fern Gully recorded the largest canopy increase, 16.1 percentage points, and 920 validated seedlings per hectare. Quartz Slope has the smallest increase, but its storm note prevents attributing that difference to treatment alone.

4. Governance and limitations

Approval state, rejected rule and abstention scope

Approval

Working paper version 1.0 was approved for internal evaluation on 9 December 2026 by the Reserve Science Lead. Public-policy adoption requires a separate review and is not implied by this approval.

Rejected rule

A draft proposed success whenever canopy rose by 12 percent. Reviewers rejected that relative rule. The approved study reports absolute percentage-point change and retains the registered paired-plot analysis.

Monitoring

Plots will be photographed each spring for four additional years. A plot is retired only after a documented boundary change; poor outcomes are not a reason for removal.

Limitations

The reserve experienced one unusually wet season. Results do not estimate wildlife abundance, carbon-credit value, wildfire probability or neighboring property impacts.

Citation

Baseline and outcome answers cite pages 2 and 3. Block questions cite the table row on page 4. Unsupported personal or financial requests require abstention.